



AP Calculus Practice Set

Writer: Jingzhe 10-8

Editor: Guoshwan 10-11

Range: AP Calc 1.1-10.3, Pre-calc Chapters 9-12

Note: (Review Guide/Review Practices/Practice Set) are UNOFFICIAL sources of reference and do NOT include or hint at questions in the actual exam!

Also, we are students, not teachers. If your answer doesn't match with ours, double check with Wolfram, Deepseek, or a teacher if you're sure you're right

I. Calculus

1. Evaluate the following limits

a) $\lim_{x \rightarrow 0} \frac{\sin(2x)}{x}$	b) $\lim_{x \rightarrow \pi} \frac{\sin(2x)}{x}$
c) $\lim_{x \rightarrow 0} \frac{1}{x}$	d) $\lim_{x \rightarrow 3} \frac{x^3 - 3x^2 - x + 3}{x^2 - 2x - 3}$
e) $\lim_{x \rightarrow 0} \frac{x^2 - 2x}{x}$	f) $\lim_{x \rightarrow 0} \frac{\sin(x^3)}{x^2}$

2. Evaluate the derivatives of the following expressions

a) $x^2 + x + 20$	b) $2x^4 - 23x + 5$
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c) $\sin(2x)$	d) $\sin(2x)\cos(x)$
e) 2^x	f) $(\cos(x^3 + \sin(2x)))^2$
g) $\sin^{-1}\frac{1}{\tan(x)}$	h) $\ln(3^x)$

3. Evaluate the following expressions

a) $\int_0^5 e^x dx$	b) $\int_0^\pi \cos(x) dx$
c) $\int_0^\infty \frac{1}{x^3} dx$	d) $\int_3^8 x^2 \cos(x) dx$

e) $\int_0^{10} -\frac{2x-2}{(x-3)(x+1)} dx$	f) $\int_4^2 x^2 dx$
g) $\int_0^\pi \cot x dx$	h) $\int_1^0 \sin^{-1} x dx$

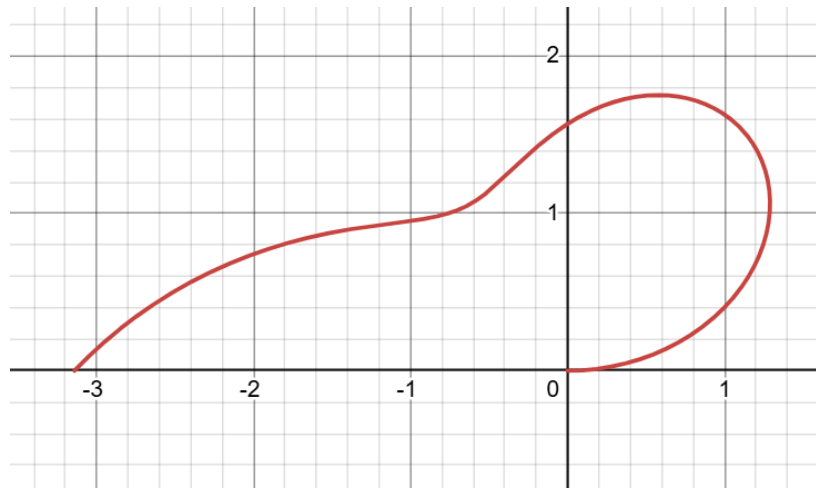
4. Let R be the region in the first and second quadrants bounded by the graph of $y = \frac{20}{1+x^2}$ and the graph of $y = 2$.

(a) Find the area of R.

(b) Find the volume of the solid generated when R is rotated about the x-axis.

(c) The region R is the base of a solid. For this solid, the cross sections perpendicular to the x-axis are semicircles. Find the volume of this solid.

5. The curve below is drawn in the xy-plane and is described by the



equation in polar coordinates $r = \theta + \sin(2\theta)$ for $0 \leq \theta \leq \pi$, where r is measured in meters and θ is measured in radians. The derivative of r with respect to θ is given by $\frac{dr}{d\theta} = 1 + 2\cos(2\theta)$.

a) Find the area bounded by the curve and the x-axis.

(b) Find the angle θ that corresponds to the point on the curve with x-coordinate -2.

(c) For $\frac{\pi}{3} < \theta < \frac{2\pi}{3}$, $\frac{dr}{d\theta}$ is negative. What does this say about r ? What does this say about the curve?

(d) Find the value of θ in the interval $0 \leq \theta \leq \frac{\pi}{2}$ that corresponds to the point on the curve in the first quadrant with greatest distance from the origin. Justify your answer.

6. Let $f(x)$ be the function given by $f(x) = \sin(5x + \frac{\pi}{4})$, and let $P(x)$ be the third-degree Taylor polynomial for $f(x)$ about $x = 0$.

(a) Find $P(x)$.

(b) Find the coefficient of x^{22} in the Taylor series for $f(x)$ about $x = 0$.

(c) Use the Lagrange error bound to show that

$$\left| f\left(\frac{1}{10}\right) - P\left(\frac{1}{10}\right) \right| < \frac{1}{100}$$

(d) Let $G(x)$ be the function given by $G(x) = \int_0^x f(t)dt$. Write the third-degree Taylor polynomial for G about $x = 0$.

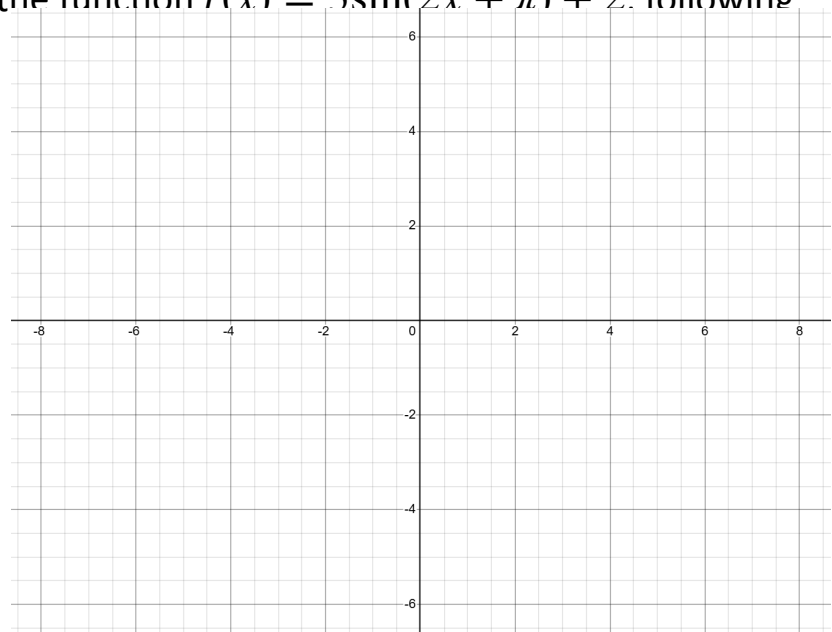
II. Pre-calculus

1. Find the values of the following expressions

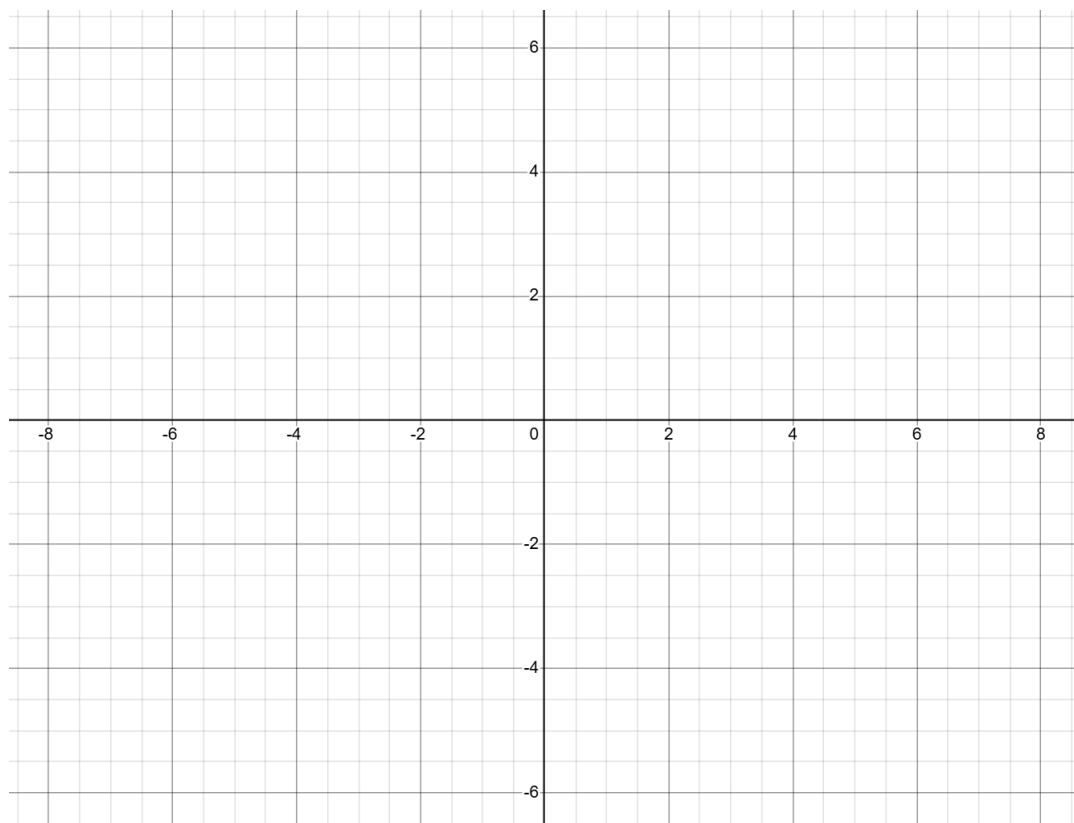
a) $\sin(\cos^{-1}(\frac{1}{2}))$	b) $\cos(\tan^{-1}(\frac{1}{\sqrt{3}}))$
c) $\sin^{-1}(\sin(\frac{2\pi}{3}))$	d) $\cos^{-1}(\cos\frac{11\pi}{6})$
e) $\cot(\tan^{-1}(1))$	f) $\cos\frac{5\pi}{2}$

2. Draw the graphs for the function $f(x) = 3\sin(2x + \pi) + 2$ following the instructions.

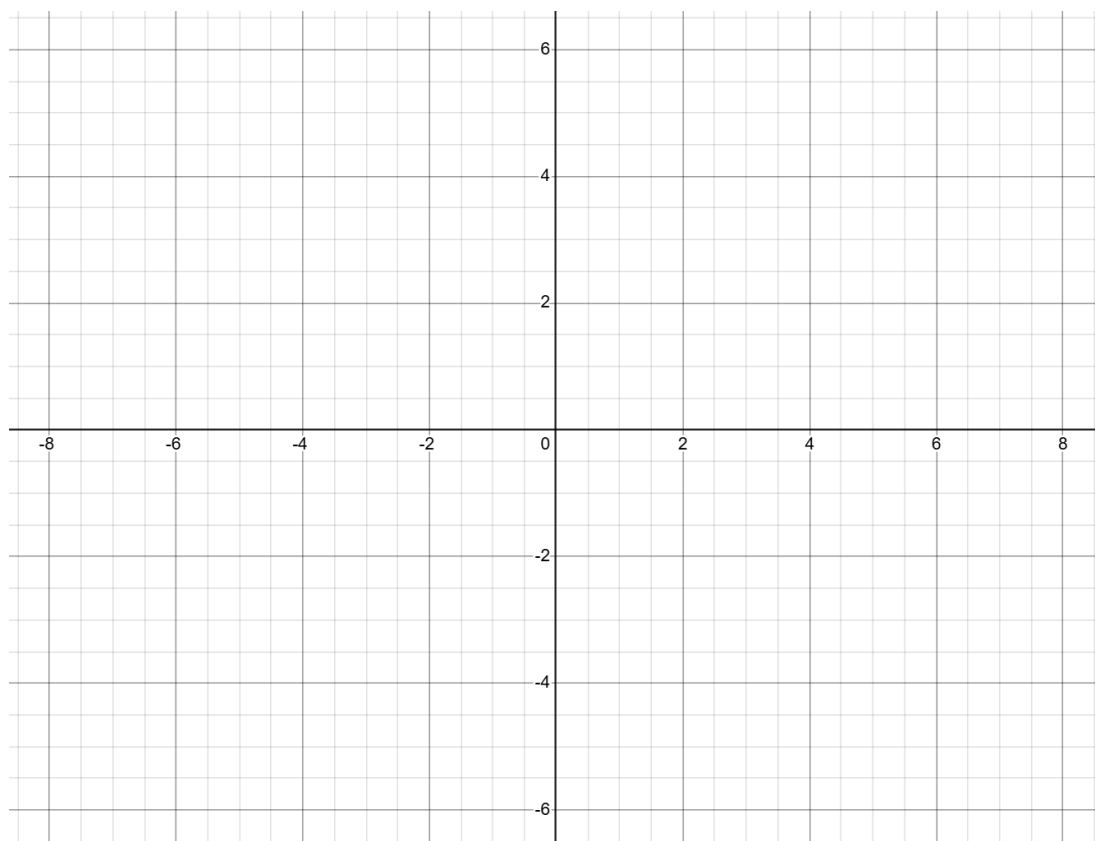
a) Draw $f(x) = \sin 2x$



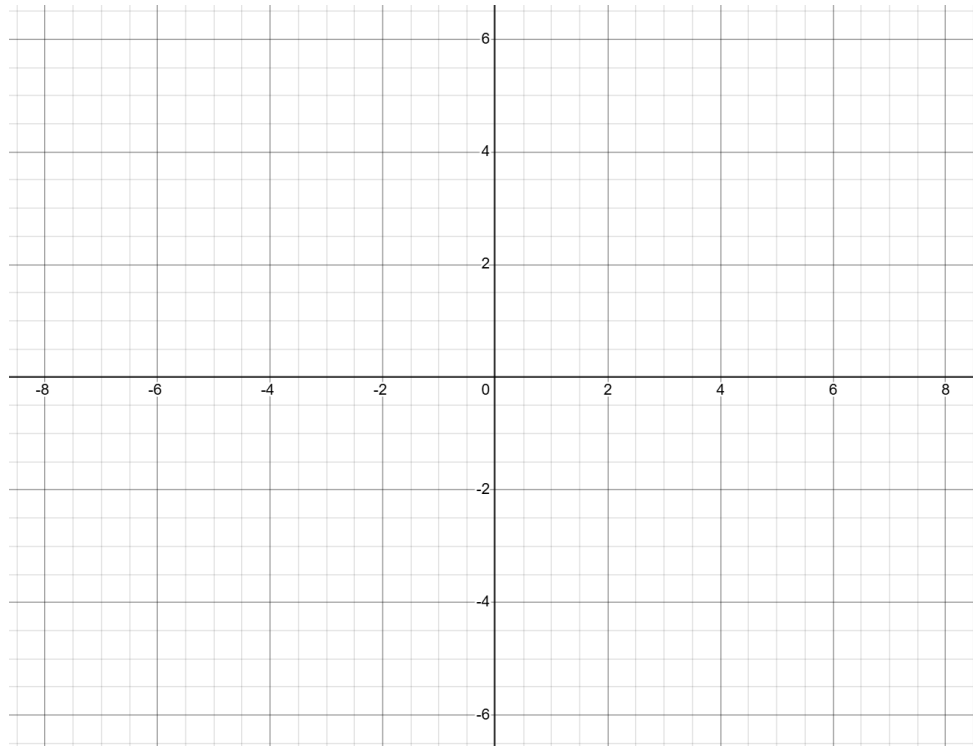
b) Draw $f(x) = \sin(2x + \pi)$



c) Draw $f(x) = 3\sin(2x + \pi)$



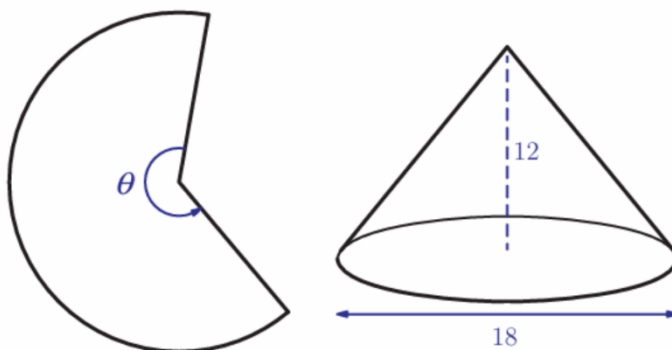
d) Draw $f(x) = 3\sin(2x + \pi) + 2$



4. A sector of a circle has perimeter $p = 12$ and angle at the center $\theta = 0.4$. Find the radius of the circle.

4. A cone is made by rolling a piece of paper as shown in the diagram. Find the value of θ .

5. Garrot stands 12 m from the angle of elevation of 52° . Find what angle of elevation does



6. $f(x) = x^3 + ax^2 + 3x + b$

The remainder when $f(x)$ is divided by $(x + 1)$ is 6. Find the remainder when $f(x)$ is divided by $(x - 1)$.

7. The polynomial $x^2 - 5x + 6$ is a factor of $2x^3 - 15x^2 + ax + b$. Find the values of a and b .